

# Foundations-Cleanup Synchronization Release for HAOS-IIP

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## Abstract

This paper freezes release 50.2 of HAOS-IIP as a small foundations-cleanup synchronization layer immediately after 50.1. The purpose is narrow and explicit. Release 50.1 synchronized the public-state comparison and the broader continuum / derivation tranche, but it still listed three bounded foundational cleanup items as open: the incidence-normalization caveat (F2), channel completeness and the no-extra-bridge rule (F3), and the Hilbert-complex upgrade of the finite-dimensional sector theorem (F4). Release 50.2 records that those three items are now closed in bounded note form. No new numerics are introduced, no disorder-threshold outputs are frozen, and no stronger continuum claim is made. The role of this paper is simply to state what changed, what did not change, and why the next clean frontier is numerical stress-testing of the already-closed theorem ladder rather than more internal foundations hygiene.

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## 1 Scope

Release 50.2 is intentionally small.

It does not add:

- new scalar numerics;
- new vector numerics;

- new disorder-threshold result bundles;
- stronger continuum claims.

It does add:

- a bounded closure of **F2**;
- a bounded closure of **F3**;
- a bounded closure of **F4**;
- spine and cross-reference updates so the theorem ladder reads continuously.

## 2 What Was Still Open in 50.1

Release 50.1 was honest about three remaining foundational cleanup items:

1. incidence normalization caveat (**F2**);
2. channel completeness and the no-extra-bridge rule (**F3**);
3. Hilbert-complex upgrade of the finite-dimensional sector theorem (**F4**).

Those were not cosmetic loose ends. They marked the points where the theorem ladder still depended on bounded caveats that had not yet been packaged as explicit standalone refinements.

## 3 What 50.2 Closes

### 3.1 **F2: canonical compatibility normalization**

The **T4** note had already proved the exact no-skip law

$$d_{k+1}Q_{k+1}^+d_k = 0.$$

What remained open was whether the familiar chain identity was only a convenient coordinate choice.

Release 50.2 closes that in bounded form by showing that the middle-grade compatibility operator itself determines a unique positive square-root weight

$$(Q_{k+1}^+)^{1/2},$$

and that the compatibility-weighted incidence maps satisfy the chain form exactly. The remaining freedom is only unitary equivalence on the active compatibility channel.

### 3.2 **F3: channel completeness and no-extra-bridge**

The **T5** note had already isolated the upward and downward incompatibility channels, but it treated the exclusion of a direct bridge operator as a minimality rule.

Release 50.2 sharpens that by identifying the bounded adjacent-grade incompatibility sector with the two-channel map

$$C_k(\omega) = (d_{k-1}^*\omega, d_k\omega),$$

then showing that every positive quadratic defect on that sector is governed by one positive Gram operator on the channel space. In that representation, an explicit bridge block is only a channel-coordinate presentation of the same positive quadratic law and disappears after canonical orthogonalization.

### 3.3 F4: Hilbert-complex upgrade

The original T6 theorem was exact in finite-dimensional inner-product spaces.

Release 50.2 upgrades that statement to Hilbert-complex language:

- without closed-range assumptions, the exact / harmonic / coexact split holds with closures on the image sectors;
- with closed-range hypotheses, one recovers the exact orthogonal direct sum

$$V_k = \text{im } d_{k-1} \oplus \mathcal{H}^k \oplus \text{im } d_k^*.$$

So the finite-dimensional theorem is now embedded in a standard closed-range Hilbert-complex form without changing the bounded interpretation of the repo.

## 4 Delta Table

| Item             | In 50.1              | In 50.2                                           |
|------------------|----------------------|---------------------------------------------------|
| F2               | Open caveat after T4 | Canonical positive-square-root normalization note |
| F3               | Open caveat after T5 | Channel-space Gram / orthogonalization note       |
| F4               | Open caveat after T6 | Closed-range Hilbert-complex upgrade note         |
| Spine status     | F2--F4 still queued  | F2--F4 marked executed in bounded form            |
| Numerics         | Unchanged            | Unchanged                                         |
| Vector threshold | Still open           | Still open                                        |

## 5 What Did Not Change

Release 50.2 does not alter the numerical authority boundary.

In particular, it does not freeze:

- any new disorder-threshold bundle;
- any new transport, gap-scan, or continuum-comparison run;
- any stronger vector-sector continuum closure claim.

So the meaning of 50.2 is not “the vector branch is now solved.” Its meaning is simply that the internal theorem-ladder caveats listed in 50.1 have now been packaged and closed in bounded note form.

## 6 Consequence for the Next Step

Because F2--F4 are now closed in bounded form, the clean next frontier is no longer theorem-ladder housekeeping. The next real stress test is numerical:

- transported active-branch identity under refinement;

- threshold behavior under bounded disorder;
- universality, effective-law, and geometry / response closure on the same derived active branch.

That is exactly where the repo should spend effort next.

## 7 Conclusion

Release 50.2 is a foundations-cleanup synchronization release, not a new-physics release. It exists so that the repository record says exactly what is now true:

- the T1--T8 ladder is still the core bounded derivation stack;
- the post-50.1 cleanup items F2--F4 are now executed in bounded form;
- the remaining frontier is numerical and continuum-facing, not internal theorem-ladder repair.

That is the whole point of 50.2.